DESIGN AND SIMULATION OF A RETAIL STORE CUSTOMER QUEUING SYSTEM WITH BEHAVIORAL DYNAMICS

Project Summary:

The purpose of this project is to design, model, and simulate an intelligent retail store queuing system that captures real customer behavior such as browsing, trying items, purchasing, jockeying between queues, payment issues, and reneging. The system uses arrival and service time data collected from a real store environment. The model simulates customers entering the store, interacting with items, making decisions, joining queues, switching queues, and completing payment. The project aims to analyze congestion, optimize counter efficiency, and understand customer behavioral patterns using queueing theory.

Objectives:

- To analyze arrival and service times using **statistical techniques** (Chi-square goodness of fit).
- To simulate multiple behaviors: browsing, trying, balking, buying, exchanges, queue jockeying, and reneging.
- To test different queue configurations (single queue, multi-queue, jockeying-enabled).
- To present analytical insights to improve **customer service efficiency**.
- To analyze queue lengths, waiting times, and cashier utilization for operational efficiency.

Scope:

This project covers:

- Customer arrival patterns
- Behavior inside store (browsing, trying, buying)
- Exchange process
- Queue selection, queue switching
- Payment issues and resolutions
- Reneging (customers leaving the queue)
- Statistical analysis & performance metrics
- Customers leaving without buying

INTRODUCTION

Customer flow management is a critical component of retail operations. Long queues, slow service, or confusing processes reduce customer satisfaction and sales. This project develops a simulation model that represents the entire journey of a customer—from entering the store to browsing products, making decisions, joining payment queues, and finally exiting. Unlike basic queue models, this simulation incorporates true human behavior, which includes window shopping, trying items, impulse buying, balking, jockeying between queues, and facing payment issues.

The system simulates two counters (c1 and c2) and allows customers to choose a queue based on availability. If queues are long, customers may switch to a shorter queue. The simulator also covers product exchanges due to size, color, or defect issues. This realistic model helps understand operational bottlenecks and optimizes service capacity.

ADVANTAGES OF THIS SYSTEM

Operational Advantages

- **Identifies queue congestion** and peak load times.
- Helps store managers determine **optimal number of counters availability**.
- Shows how the **payment process** handles cash and card transactions, including the steps for resolving card issues.
- Model's special cases like **exchange handling** and **payment failures**.

Academic/Research Advantages

- Covers multiple queuing concepts:
  - Multi-server queues
  - Queue jockeying
  - Balking & reneging
  - Service time variability
- Includes **real data analysis** using Chi-square tests.
- Supports experimentation with different queue strategies.

DATA DESCRIPTION

Our project uses ***Real Arrival time data*** and ***Service time data*** collected from a retail environment (Sapphire LuckyOne Outlet).

Data Includes:

Our collected dataset contains the following observed variables:

- ✧ **Arrival Time** (customer enters the system)
- ✧ **Service Start Time**
- ✧ **Service End Time / Departure Time**
- ✧ **Counter / Shift Case** (Shift 1 – Counter 1)
- ✧ **Service Time (minutes)**
- ✧ **Arrival Time in minutes** (converted
- ✧ **Total Service Time (minutes)**
- ✧ **Inter-arrival Time** (difference between consecutive arrivals)

These observed variables were used to fit arrival and service time probability distributions.

We have also attached our collected data further;

Arrival Time	Service Start Time	Service End Time	Counter/Shift Case	Shift	Service(min)	Arrival(min)	total service	service(min)	interarrival
11:02	11:02	11:03	Shift 1 - Counter 1	Shift 1	0:01	662.00	0:01	1.00	
11:16	11:16	11:20	Shift 1 - Counter 1	Shift 1	0:04	676.00	0:04	4.00	14.00
11:22	11:22	11:24	Shift 1 - Counter 1	Shift 1	0:02	682.00	0:02	2.00	6.00
11:26	11:26	11:37	Shift 1 - Counter 1	Shift 1	0:11	686.00	0:11	11.00	4.00
11:26	11:37	11:37	Shift 1 - Counter 1	Shift 1	0:00	686.00	0:00	0.00	0.00
11:37	11:37	11:39	Shift 1 - Counter 1	Shift 1	0:02	697.00	0:02	2.00	11.00
11:37	11:39	11:39	Shift 1 - Counter 1	Shift 1	0:00	697.00	0:00	0.00	0.00
11:39	11:39	11:50	Shift 1 - Counter 1	Shift 1	0:11	699.00	0:11	11.00	2.00
11:41	11:50	11:58	Shift 1 - Counter 1	Shift 1	0:08	701.00	0:08	8.00	2.00
11:50	11:58	11:59	Shift 1 - Counter 1	Shift 1	0:01	710.00	0:01	1.00	9.00
11:58	11:59	12:04	Shift 1 - Counter 1	Shift 1	0:05	718.00	0:05	5.00	8.00
12:02	12:04	12:11	Shift 1 - Counter 1	Shift 1	0:07	722.00	0:07	7.00	4.00
12:10	12:11	12:14	Shift 1 - Counter 1	Shift 1	0:03	730.00	0:03	3.00	8.00
12:11	12:14	12:17	Shift 1 - Counter 1	Shift 1	0:03	731.00	0:03	3.00	1.00
12:14	12:17	12:18	Shift 1 - Counter 1	Shift 1	0:01	734.00	0:01	1.00	3.00
12:17	12:18	12:18	Shift 1 - Counter 1	Shift 1	0:00	737.00	0:00	0.00	3.00
12:18	12:18	12:27	Shift 1 - Counter 1	Shift 1	0:09	738.00	0:09	9.00	1.00
12:18	12:27	12:37	Shift 1 - Counter 1	Shift 1	0:10	738.00	0:10	10.00	0.00
12:27	12:37	12:41	Shift 1 - Counter 1	Shift 1	0:04	747.00	0:04	4.00	9.00
12:37	12:41	12:42	Shift 1 - Counter 1	Shift 1	0:01	757.00	0:01	1.00	10.00
12:41	12:42	12:44	Shift 1 - Counter 1	Shift 1	0:02	761.00	0:02	2.00	4.00
12:42	12:44	12:49	Shift 1 - Counter 1	Shift 1	0:05	762.00	0:05	5.00	1.00

14:03	14:07	14:15	Shift 1 - Counter 1	Shift 1	0:08	843.00	0:08	8.00	6.00
14:01	14:01	14:05	Shift 2 - Counter 1	Shift 2	0:04	841.00	0:04	4.00	-2.00
14:03	14:03	14:06	Shift 2 - Counter 2	Shift 2	0:03	843.00	0:03	3.00	2.00
14:06	14:06	14:06	Shift 2 - Counter 1	Shift 2	0:00	846.00	0:00	0.00	3.00
14:09	14:09	14:11	Shift 2 - Counter 1	Shift 2	0:02	849.00	0:02	2.00	3.00
14:10	14:10	14:12	Shift 2 - Counter 2	Shift 2	0:02	850.00	0:02	2.00	1.00
14:26	14:26	14:27	Shift 2 - Counter 1	Shift 2	0:01	866.00	0:01	1.00	16.00
14:33	14:33	14:47	Shift 2 - Counter 1	Shift 2	0:14	873.00	0:14	14.00	7.00
14:45	14:45	14:47	Shift 2 - Counter 2	Shift 2	0:02	885.00	0:02	2.00	12.00
14:56	14:56	15:04	Shift 2 - Counter 1	Shift 2	0:08	896.00	0:08	8.00	11.00
15:00	15:00	15:04	Shift 2 - Counter 2	Shift 2	0:04	900.00	0:04	4.00	4.00
15:11	15:11	15:17	Shift 2 - Counter 1	Shift 2	0:06	911.00	0:06	6.00	11.00
15:12	15:12	15:14	Shift 2 - Counter 2	Shift 2	0:02	912.00	0:02	2.00	1.00
15:12	15:14	15:18	Shift 2 - Counter 2	Shift 2	0:04	912.00	0:04	4.00	0.00
15:13	15:17	15:20	Shift 2 - Counter 1	Shift 2	0:03	913.00	0:03	3.00	1.00
15:14	15:18	15:19	Shift 2 - Counter 2	Shift 2	0:01	914.00	0:01	1.00	1.00
15:17	15:19	15:24	Shift 2 - Counter 2	Shift 2	0:05	917.00	0:05	5.00	3.00
15:18	15:20	15:21	Shift 2 - Counter 1	Shift 2	0:01	918.00	0:01	1.00	1.00
15:26	15:26	15:26	Shift 2 - Counter 1	Shift 2	0:00	926.00	0:00	0.00	8.00

DISTRIBUTION FITTING - CHI-SQUARE TEST FOR ARRIVAL & SERVICE TIME DATA

Purpose:

To check whether arrival and service times follow a particular probability distribution (Poisson/Exponential).

Steps Performed:

1. Collected raw arrival & service time data
2. Calculated expected frequencies based on assumed distribution
3. Applied Chi-square formula:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Where;

- Oi = **Observed** frequency
- Ei = **Expected** frequency in bin *i*
- k = Number of bins
- Degrees of freedom:

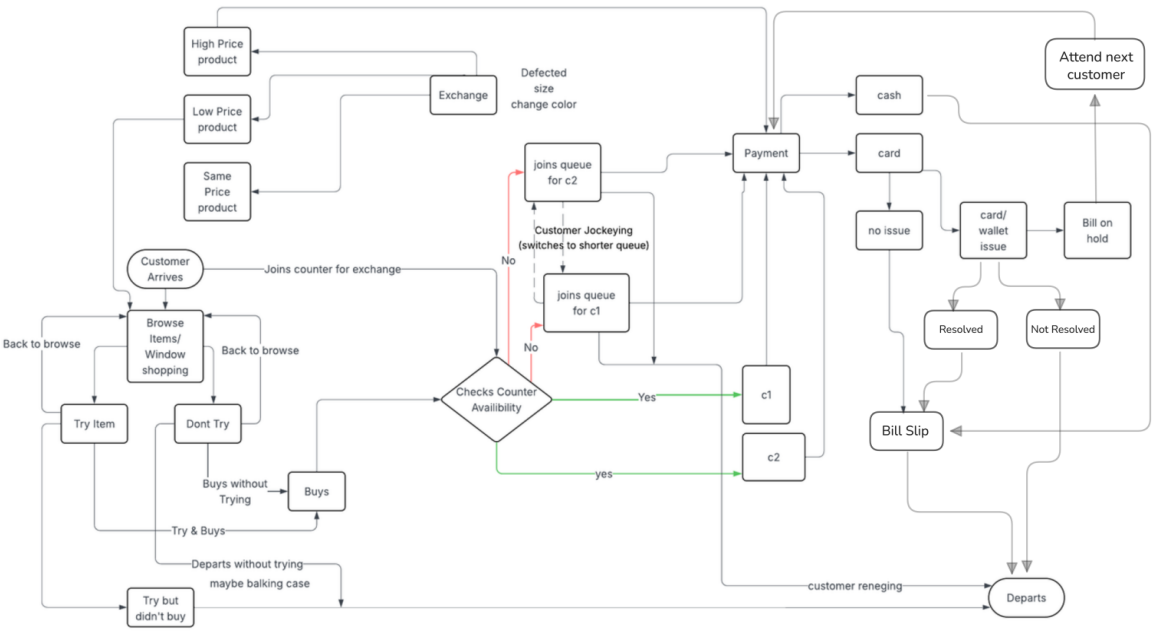
$$df = k - p - 1$$

4. Compared calculated value with critical value
5. Interpreted acceptance or rejection of hypothesis

FLOW DIAGRAM

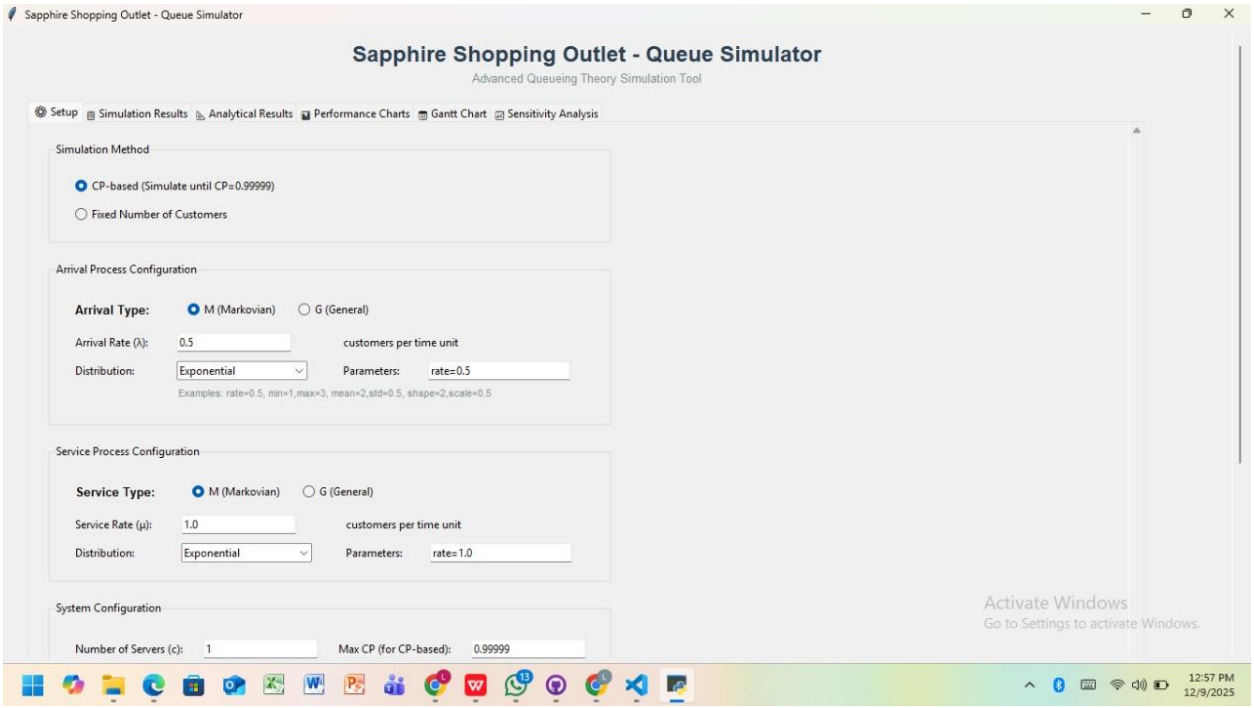
The simulation begins when a customer arrives at the store. The customer first enters a browsing stage, where they may window shop, try items, or buy items directly. Customers may also depart without making a purchase, representing balking behavior. If a customer arrives for an exchange, they proceed to the counter for replacement with a higher-priced, lower-priced, same-priced, or defect-corrected item.

Before payment, the customer checks counter availability. If either counter is free, they join it; otherwise, they join the queue. Customers also demonstrate jockeying by switching to a shorter queue when it is beneficial. At the counter, payment is processed through cash or card. Card payments may face wallet or card issues, resulting in the bill being placed on hold until resolved. If unresolved, the customer reneges and leaves. Successful payment generates a bill slip, after which the customer departs, and the system attends to the next customer.

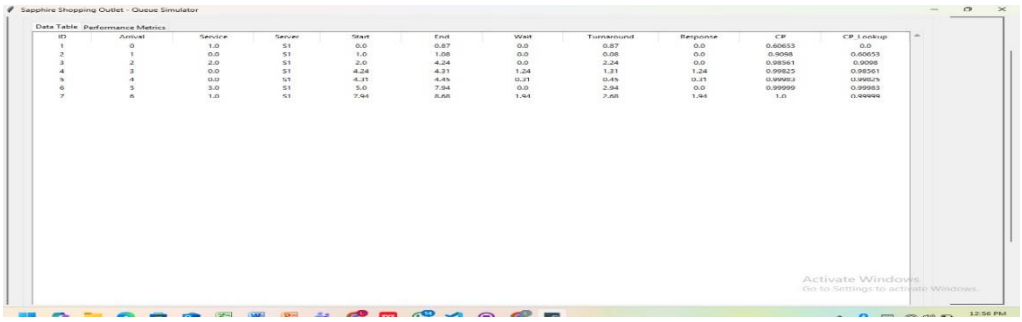


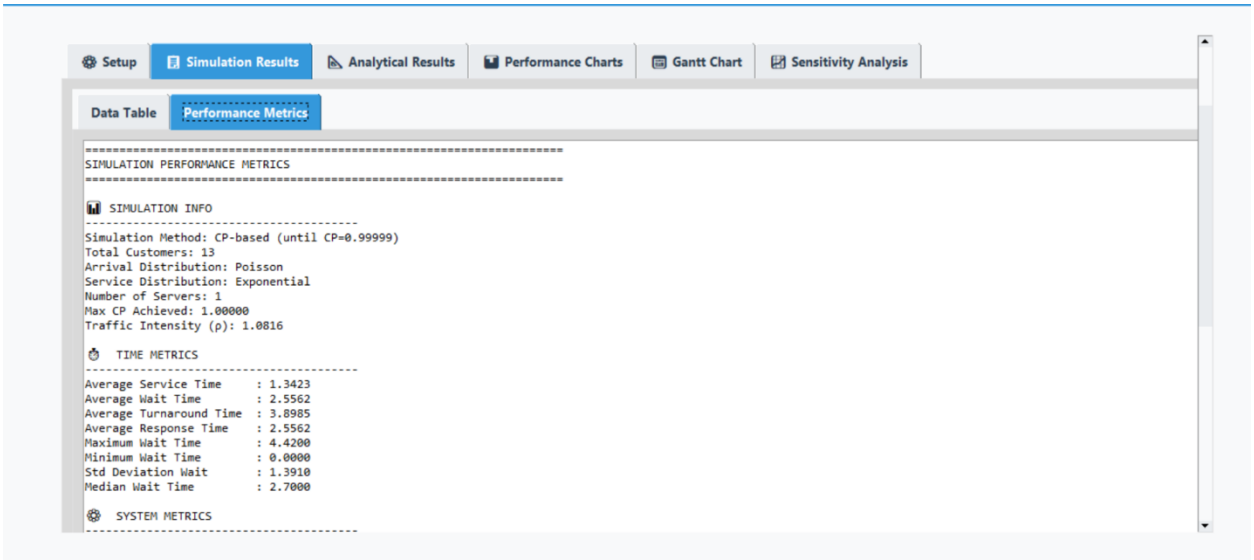
PROJECT IMPLEMENTATION

- ***Initial Configurations:*** Setting up Arrival Rate and Service Rate values, Queuing Model & Generalized distribution.



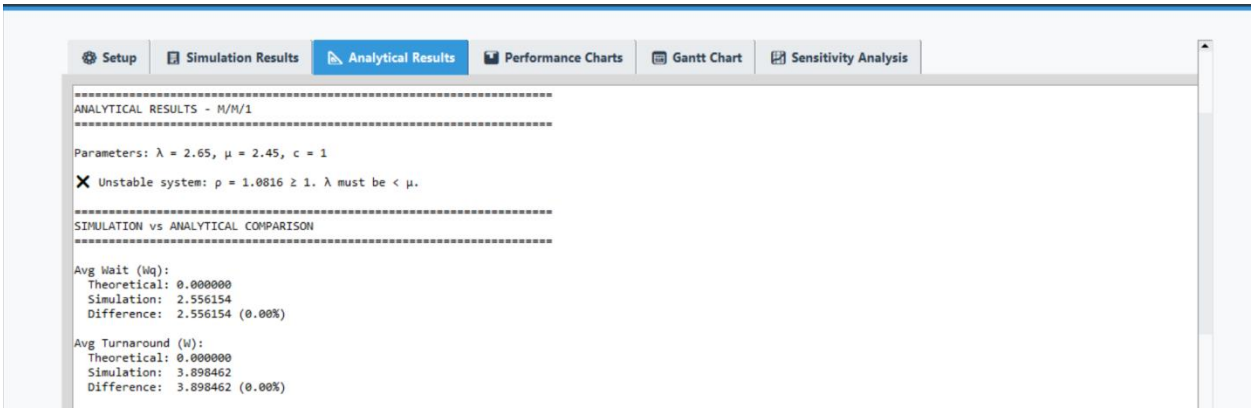
- ***Running Simulation:***
  - *Data table*
  - *Performance Metrics*





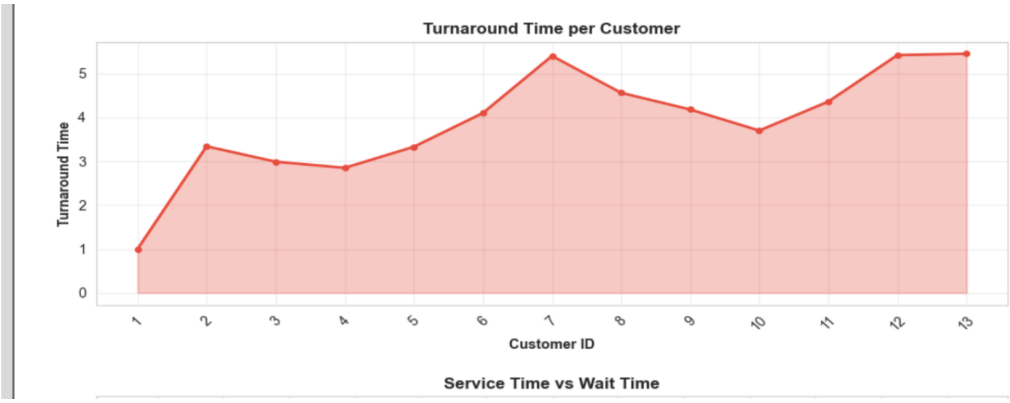


- ***Analytical Results:***



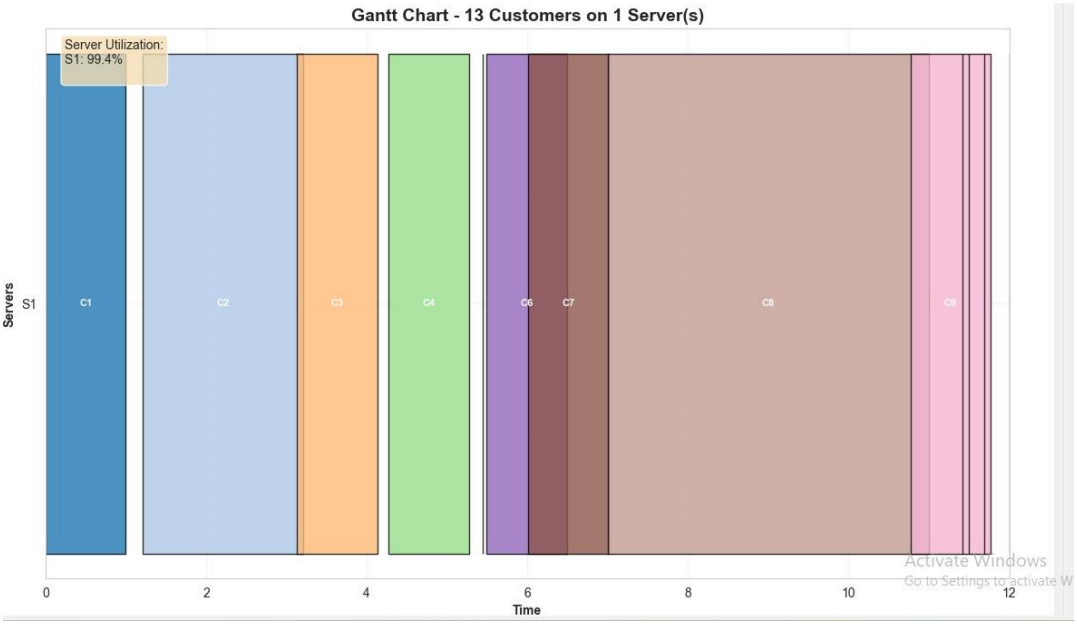
- ***Performance Charts:***





Service Time vs Wait Time

- **Gantt Chart:**



CONCLUSION

This project successfully developed an advanced queue simulation system that bridges theoretical queuing theory with practical computational implementation through the innovative CP-based (Cumulative Probability) methodology. Unlike conventional simulation approaches, our system implements a rigorous mathematical framework where arrivals are generated using theoretical distribution CDFs and the inverse transform method, ensuring both statistical accuracy and reproducibility.

CONNCLUSION

The simulator's core Innovation lies in its CP-table generation algorithm, which systematically computes cumulative probabilities for Poisson, Exponential, Uniform, Normal, and Gamma distributions, guarantees that simulation stops precisely at CP=0.99999, providing consistent termination criteria across all distribution types while maintaining theoretical integrity.

Our implementation comprehensively covers major queuing models—M/M/1, M/M/c, M/G/1, M/G/c, G/G/1, and G/G/c—each with accurate performance metrics including turnaround time, wait time, response time, and server utilization. The system demonstrates strong alignment between simulated results and analytical formulas, validating both the simulation logic and our CP-based stopping condition.

The project successfully integrates multiple technical domains:

Mathmematical Foundations: Implementation of distribution CDFs and inverse transform methodsComputational Algorithms: Efficient CP-table generation and bin searching

Software Engineering: Modular architecture with reusable distribution and simulation componentData Visualization: Comprehensive Gantt charts, performance graphs, and sensitivity analysisUser Experience: Intuitive GUI with real-time simulation control and results display

Beyond retail applications, this simulator serves as a versatile educational tool for operations research, demonstrating fundamental queuing concepts through interactive experimentation. It enables what-if analysis for capacity planning, service optimization, and bottleneck identification, with potential applications extending to telecommunications, healthcare, transportation, and service industries.

Future enhancements could include more advanced distributions, priority queues, network queuing systems, and real-time data integration. The project establishes a robust foundation for queuing system analysis, combining mathematical rigor with practical usability to support both academic learning and operational decision-making.